21 Genomes and Their Evolution

KEY CONCEPTS

Study Tip

Write genomic analysis questions: As you go through the chapter, make a table of genomic analyses that are discussed and write down some questions they can address. You can include questions not covered in the text.

Genomic projects and websites	Questions that can be addressed
Human Genome Project	How much DNA is in the human genome? How much of the human genome codes for proteins? What functions are carried out by other regions of the genome?
ENCODE project	

Go to Mastering Biology

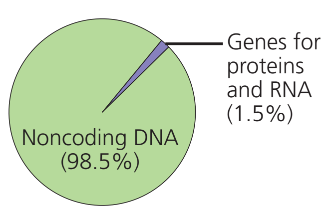
For Students (in eText and Study Area)
- Get Ready for Chapter 21
- HHMI Animation: Shotgun Sequencing
- Animation: The Human Genome Project: The Genes on Human Chromosome 17

For Instructors to Assign (in Item Library)
- Tutorial: Scientific Skills Exercise: Reading an Amino Acid Sequence Identity Table
- Tutorial: Using BLAST: What Can a Protein Sequence Reveal About Cancer?

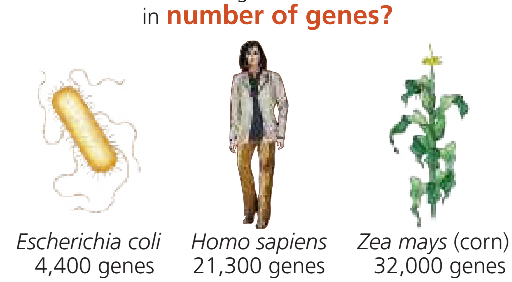
Figure 21.1 The elephant shark (*Callorhinchus milii*) looks vaguely prehistoric and has been called a "living fossil." In fact, it has the slowest evolving genome of any vertebrate sequenced so far. Comparing rates of genome change in different species provides insights into the evolutionary past.

What are some questions that can be explored by sequencing and comparing genomes?

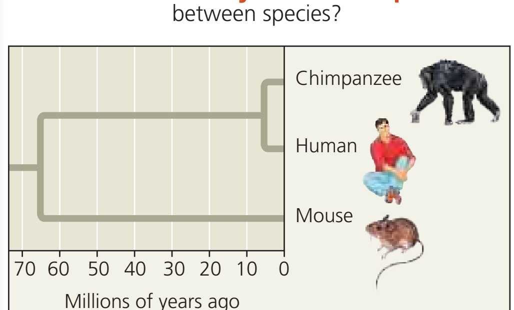
What are the **functions** of the human genome?



How do genomes differ in **number of genes?**



What do gene sequences tell us about **evolutionary relationships** between species?



How do genomes **evolve** over time?